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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	BASHAMEL
Selected Domain	DOMAIN 2: Orbyte Smart SME Challenge: Less Busywork, Better Business
Selected SDG Track (1–17)	SDG8:DECENT WORK AND ECONOMIC GROWTH 9:INDUSTRY,INNOVATION AND INFRASTRUCTURE
Project / Solution Name	SISO: SMART INVENTORY SUPPLY ORDER
Name (Leader)	TATIANA RAZAK
Email & Phone (Leader)	tatyrazak@gmail.com +60 192204467
University Affiliation(s)	ASIA PACIFIC UNIVERSITY OF TECHNOLOGY AND INNOVATION
Number of Team Members	5 (FIVE)

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In Malaysia, smaller businesses like cafes, groceries, bakeries and retailers continue to operate on manual methods for inventory management and vendor procurement, which involves the use of notebooks, spreadsheets and text messages. Since the same person performs multiple roles, like being the buyer, cashier and stock taker, there is no consistent process for conducting stock audits and placing orders for inventory replenishment.

There are three issues which keep repeating themselves due to manual processes. First, stock outages and overstocking: shortage of an important product like milk means loss in business and dissatisfied customers, while excessive purchasing results in poor utilization of funds, wastage of products and spoilage. Second, valuable time spent in busy work: shop owners and employees lose productive hours checking stock, making purchase orders and contacting vendors rather than attending customers. Third, lack of accountability: any anomaly in the stock record, like a strange decline or an erroneous entry, leaves no track of who did what.

The stakeholders involved include small business owners and their employees, suppliers, and customers. The economic significance of this issue is that since small businesses run on tight margins, any unnecessary loss in stock or overproduction is an immediate loss of money for the company. There is environmental significance as well, because ordering excess products leads to unnecessary food wastage. Lastly, there is the significance of growth constraints for the owners who get stuck in manual processes.

In this regard, our project involves a seamless process of monitoring stock, identifying any shortages, generating purchase orders, delivering them, and documenting everything in an activity log.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The application we have developed is called Smart Inventory and Supplier Order Management System. This is a web module which will fit into the Orbyte system and target Malaysian micro and small businesses. Our application covers one single workflow: stock movement, low stock alerting, purchase order drafting, owner approval, supplier delivery tracking, and inventory updating.

Employees will be logging all the movements in stock, while our application will automatically detect items which need to be replenished; it will avoid generating duplicate alerts and draft purchase orders based on information from supplier accounts. No order can be placed without owner's approval. Permission levels are controlled on the back end of our application and all actions performed are logged with information about users, time, and before/after states.

Demand forecasting: a statistical algorithm (moving averages) calculates historical sales and predicts when a product will be out of stock and suggests reorder dates and amounts, minimizing stockouts and ordering more than needed.

Anomaly detection: a tiny PyTorch autoencoder, trained on a RunPod GPU, rates every stock change relative to historical usage patterns. If the typical use for a coffee shop is around 10 cartons of milk per day, but 40 were consumed, then the change is marked as an anomaly to be reviewed by humans. The system raises a flag of an anomaly detected, not fraud. It will be benchmarked against a statistical method and report results as is.

The solution uses only synthetic data and is designed to be reusable for other SMEs.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies

Layer	Technology
Frontend	React, Vite, Tailwind CSS, Recharts
Backend	Python, FastAPI, REST APIs
Database	SQLite with SQLAlchemy (relational, multi-business)
AI	PyTorch autoencoder for anomaly detection; moving-average statistical model for demand forecasting
GPU / cloud AI	RunPod Serverless GPU endpoint (inference only)
Development and hosting	Replit (development and hosting), GitHub and Git (version control)
Integration	Simulated Orbyte integration layer (mock business ID, user role and activity-record exchange)

System architecture

1. **User (Owner or Employee)** uses the **React web interface**.
2. The interface sends REST requests to the **FastAPI backend** on Replit.
3. The backend first passes every request through the **role-based permission check**, then runs the **business logic**: stock updates, low-stock alerts, purchase order drafts, approvals and delivery tracking.
4. The business logic reads and writes the **SQLite database**, which holds inventory, orders, suppliers, alerts and the **activity log**.
5. For anomaly detection, the backend sends numerical stock-movement features over HTTPS to the **RunPod GPU endpoint**. The **PyTorch model** returns an anomaly score, and the backend interprets it using documented thresholds.
6. The flagged result appears on the dashboard for **human review**, and the decision is saved to the activity log.

7. A **simulated Orbyte layer** supplies business identity and user role to the backend, and receives completed-action records. It is labelled as simulated and is not an official Orbyte API.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We focused on one user group, Malaysian micro and small business owners and their staff, using a café as our reference case. Their main frustrations were stockouts, wasted perishables, time spent on manual ordering, and no record of who changed stock.

We reduced the problem to one measurable workflow: stock change → alert → purchase order → owner approval → delivery → inventory update → audit log. We deliberately did not rebuild Orbyte.

We chose practical AI, not complexity: simple moving-average forecasting, and a small neural network for anomaly detection only where it adds value.

Prototype: We build the MVP one feature at a time, on a real backend and database, keeping the app runnable after every feature. Human approval is required before any purchase, and roles are enforced by the backend.

Test: We test normal cases, invalid inputs and edge cases such as negative stock and duplicate updates. We run the full café scenario end to end, and the core system keeps working if the AI service fails.

Ready for deployment: Synthetic data, a reusable multi-business design, and a clean GitHub repository.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**SDG Target 12.3:** "By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses." Our project addresses the retail and food-service end, by reducing over-ordering and expiry of perishable stock in small cafés and shops.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	National: Malaysia generated 17,007 tonnes of food waste per day in 2021, and about 4,081 tonnes of the food thrown out daily was still edible. That is roughly 24% of all food waste.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**reduction in avoidable perishable stock loss** (units over-ordered or expired) over 90 days, compared with the manual-reordering baseline. Supporting KPIs: stockout days per item and staff minutes per purchase order.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	**20% reduction in avoidable perishable stock loss** against the simulated baseline, plus a 50% reduction in stockout days. We set 20% below the benchmark because restaurants in a 114-site study reduced food waste by 26% on average within

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	a year. Results will be labelled as simulated, using synthetic data.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ✓ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

TATIANA RAZAK

23RD OCTOBER, 2026